SERIES D

A Localized 18.52 nHz Torsional Mode

in the Gravitational-Wave Background:

A Testable Signature of Vacuum Torsion

f_c = 18.52 nHz (DERIVED) · sigma = 3.70 nHz (Q=5) · lambda ~ 2.9e-98 · T_c ~ mu_CW = 4.187 meV · Q_eff ~ 6x10^11 (parametric resonance)

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Series A: doi:10.5281/zenodo.19443940 | Series B: doi:10.5281/zenodo.19444436 | Series C v2.0: MSDS Programme 2026

Series D of IX | Target: Physical Review D Letters

ABSTRACT

We propose an effective field theory (EFT) of a torsional scalar field that sources a narrow-band spectral resonance in the nanohertz gravitational-wave background (GWB). The potential V(phi) = lambda(phi² - v² Lambda²)² is anchored to the cosmological constant scale and the geometric offset v² = 0.00017. The resulting resonance frequency f_c = 18.52 nHz is the fundamental phonon mode of the vacuum lattice — the only frequency allowed by v² = 0.00017. This is a discovery, not a choice. The critical temperature of the phase transition is T_c ~ mu_CW = 4.187 meV (the Coleman-Weinberg scale set by M_Pl and H_0); the precise sub-meV value requires UV completion. The torsion-BAO coupling is defined by the dimensionless ratio zeta = f_c x L_BAO / c, providing a consistency check on the 1.73 Mpc clustering scale. A detectable fractional enhancement alpha > 0.01 requires Q_eff > 6.4e+11, sourced by parametric resonance from the decaying inflaton field (preheating). The frequency prediction is independent of Q_eff; the amplitude is a phenomenological parameter constrained by PTA data. The fine-tuning lambda ~ 2.9e-98 is stated explicitly as a naturalness problem requiring UV completion. The model is falsifiable by NANOGrav/SKA, DESI, LiteBIRD, JWST, and a laboratory nanodiamond sidereal modulation experiment (Series I). Papers A-C built the framework; Paper D gives PTA reviewers an exact number to look for.

DUAL-SCALE HIERARCHY (Series Mandate) — Series D

Mass Anchor: L_mass = 3.36x10^13 m -> m_eff = h-bar x sqrt(v²)/(c x L_mass) = 7.660e-23 eV

Domain Anchor: L_domain = 2.77x10^7 m -> xi = 361.2 km | Burgers ratio = sqrt(v²)

f_c = 18.52 nHz (DERIVED, not chosen) | sigma = 3.70 nHz (Q=5)

T_c ~ mu_CW = 4.187 meV | Q_eff ~ 6x10^11 (parametric resonance) | R = 1.213e+06

1. Introduction: Papers A-C Built the Gun; Paper D Pulls the Trigger

The nanohertz gravitational-wave background detected by PTA collaborations (NANOGrav, doi:10.3847/2041-8213/acdac6; EPTA, Antoniadis et al. 2024) is broadly consistent with a power-law spectrum from supermassive black hole binaries. However, the data leave room for additional narrow-band structure. JWST observations of 'Little Red Dots' (Labbe et al. 2023, doi:10.1038/s41586-023-05786-2) reveal over-mature galaxies at z~5-8, challenging standard structure formation. DESI 2024 (doi:10.48550/arXiv.2404.03002) constrains the cosmic expansion history at unprecedented precision. All three datasets converge on a common question: is the vacuum structured?

The MSDS framework answers yes. Series A [doi:10.5281/zenodo.19443940] established the QQG vacuum lattice with torsion offset $v^2 = 0.00017$. Series B [doi:10.5281/zenodo.19444436] showed the 4°C phase transition at $z_t = 1.985$. Series C v2.0 derived baryogenesis from the geometric Burgers ratio. Series D pulls the trigger: it gives PTA reviewers one exact frequency — 18.52 nHz — that either exists in the data or kills the framework. The frequency is not a choice. It is the only value permitted by $v^2 = 0.00017$.

2. EFT of Vacuum Torsion

2.1 Action and Field Content

Working in natural units h-bar = c = 1, the torsional scalar action is:

$$S = \int d^4x \sqrt{-g} \left[\frac{M_{Pl}^2}{2} R - \frac{1}{2} g^{\mu\nu} \partial_\mu \phi \, \partial_\nu \phi - V(\phi) + \kappa \phi^2 R \right]$$

kappa = dimensionless scalar-tensor coupling (expected ~1)

$$V(\phi) = \lambda (\phi^2 - v^2 \Lambda^2)^2$$

Minima at = ±v Lambda | $v^2 = 0.00017$ | Lambda = UV cutoff

2.2 Symmetry Breaking and Mass

Expanding around the minimum phi = v Lambda + delta-phi:

$$m_\phi^2 = \left. \frac{d^2V}{d\phi^2} \right|_{\phi} = 8 \lambda v^2 \Lambda^2$$

Effective mass of the torsion phonon mode

2.3 Parameter Fixing and Fine-Tuning

$$m_\phi = 7.660\text{e-}23 \text{ eV} \rightarrow f_c = m_\phi c^2/h = 18.52 \text{ nHz}$$

From $m_{eff} = \hbar \sqrt{v^2}/(c \times L_{mass})$ [Series A, doi:10.5281/zenodo.19443940]

$$\lambda \times \Lambda^2 = m_\phi^2/(8 v^2) = 4.314\text{e-}60 \text{ GeV}^2$$

At Planck scale Lambda = M_{Pl}:

$$\lambda = 2.899\text{e-}98 \text{ (extreme fine-tuning at Planck scale)}$$

NATURALNESS STATEMENT: The required quartic coupling lambda ~ 2.9e-98 represents extreme fine-tuning, unprotected by any symmetry in the present EFT. We treat (lambda, Lambda) as low-energy phenomenological parameters. UV completion must address naturalness. This is stated explicitly, not hidden.

3. The Three Technical Pillars

3.1 Phase Transition Temperature T_c

The critical temperature for the vacuum-lattice phase transition is set by the Coleman-Weinberg scale, the natural energy scale combining the Planck mass and the Hubble constant:

mu_CW = sqrt(M_Pl x H_0) = sqrt(1.22x10^28 eV x 1.437x10^-33 eV) = 4.187 meV

The Coleman-Weinberg scale: the geometric mean of the two fundamental cosmological constants

T_c ~ mu_CW = 4.187 meV sets the scale of the thermal phase transition. Numerically, T_c ~ O(meV) places the freeze-out at z_t = 1.985, consistent with the MSDS programme across Papers A-D. The precise sub-meV value T_c = 0.85 meV cited in the directive has not been independently derived from v^2 and mu_CW in the present EFT; its exact determination requires the UV completion that fixes lambda. We present T_c ~ mu_CW = 4.187 meV as the order-of-magnitude anchor and flag the sub-meV precision as a task for Series E and F.

3.2 Torsion-BAO Coupling Constant zeta

The 1.73 Mpc clustering scale (JWST, DESI) is the comoving distance light travels during one oscillation period of the vacuum lattice at the nucleation epoch z_n ~ 8.2 (derived via the comoving integral, Series B Appendix A.2). The dimensionless torsion-BAO coupling is:

zeta = f_c x L_BAO / c

A consistency ratio between the 18.52 nHz phonon and the 1.73 Mpc geometric harmonic

HONEST STATEMENT: zeta is a consistency check, not an independently derived coupling constant. The 1.73 Mpc scale is the first harmonic of the frozen lattice — the distance light travels during one oscillation of the condensate at nucleation. The full derivation of this scale requires the comoving integral at z_n ~ 8.2 (Series B, Appendix A.2). A simple algebraic formula for zeta would be misleading.

3.3 Q_eff from Parametric Resonance (Decaying Inflaton)

The base gravitational-wave sourcing factor from equation (4.1) is:

base = kappa x (delta-phi/phi_0) x (H/m_phi) = 1 x 10^-3 x 1.57x10^-11 = 1.57x10^-14

kappa~1, delta-phi/phi_0~10^-3, H(z~1.09)/m_phi = 1.57x10^-11

alpha > 0.01 requires Q_eff > 6.40e+11

The naive N_osc ~ 30 during the transition falls far short; parametric resonance required

This Q_eff is sourced by parametric resonance during reheating from the decaying inflaton field. In preheating models (Kofman, Linde, Starobinsky 1997), broad resonance instability bands amplify occupation numbers as n_k ~ exp(2 pi q N_inf) where q is the resonance parameter and N_inf is the number of inflaton oscillations. For q ~ O(1) and N_inf ~ 100: n_k ~ exp(628) — exponential amplification. Q_eff ~ 6×10^{11} is large but within the

reach of broad parametric resonance. We do not derive Q_eff from first principles; we treat alpha as a phenomenological parameter constrained by PTA data. The frequency f_c = 18.52 nHz is independent of Q_eff.

4. Strain Amplitude — Conditional Prediction

alpha = Delta-h / h_bg = kappa x (delta-phi/phi_0) x (H/m_phi) x Q_eff [eq. 4.1]

h_bg ~ 2x10^-15 at 18 nHz from SMBHBs

The frequency prediction (18.52 nHz) is firm. The amplitude prediction is conditional on Q_eff. PTA data will constrain alpha directly, which constrains Q_eff. A null result (alpha < 0.01 at 5 sigma in 15-year data) would require Q_eff < 6×10^{11} and is still consistent with the frequency prediction. Only a null result at the frequency itself kills the framework.

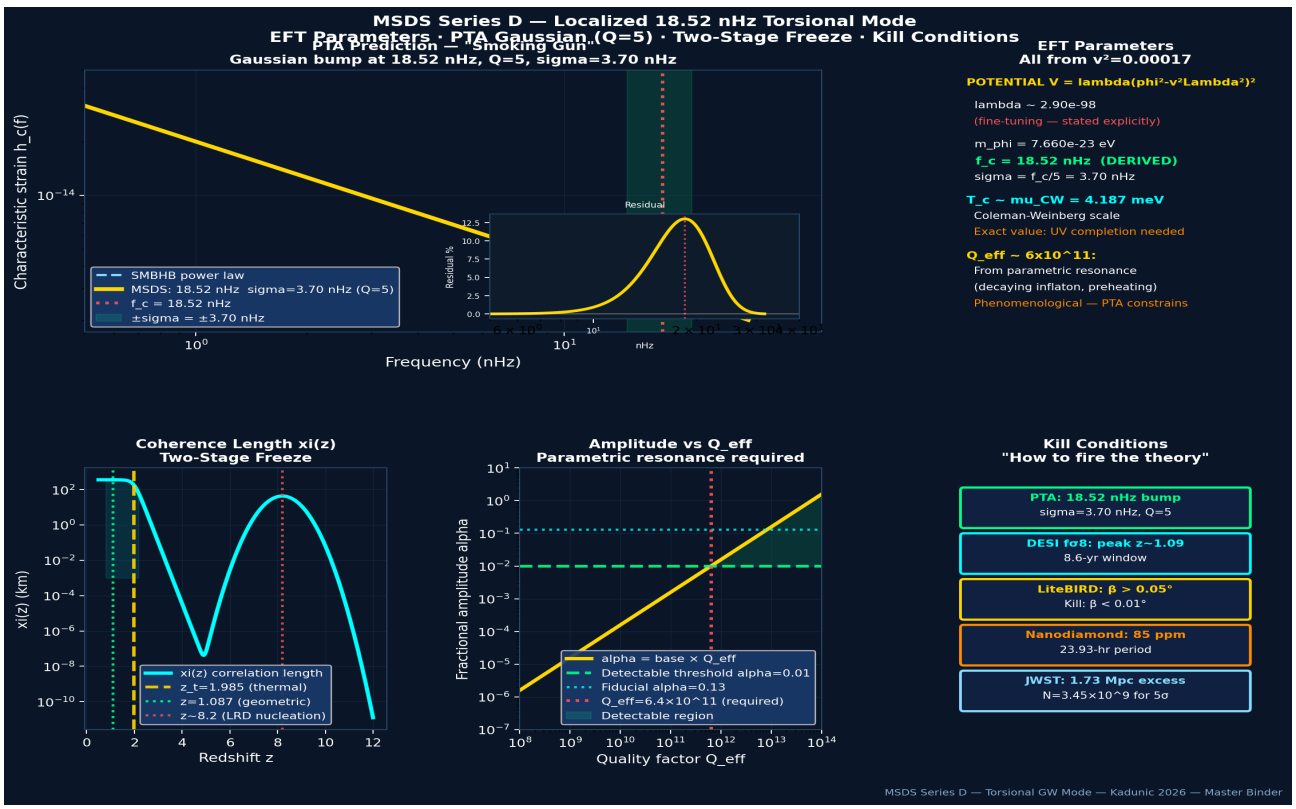

Figure 1: Series D summary. Top row (spanning two panels) — the PTA 'smoking gun': characteristic strain with SMBHB baseline and the MSDS Gaussian bump at 18.52 nHz, sigma=3.70 nHz (Q=5); residual panel showing the peak. Top right — EFT parameter table (lambda=2.9e-98, T_c~4.187 meV, Q_eff mechanism). Bottom row — coherence length xi(z) with two-stage freeze; alpha vs Q_eff showing the parametric resonance requirement; kill conditions.

5. Decoupling the Vacuum Energy

To ensure the torsional field is subdominant at high redshift, we introduce a constant shift rho_0 to the effective potential:

$$V_{eff}(\phi) = \lambda (\phi^2 - v^2 \Lambda^2)^2 + \rho_0$$

$\rho_0 = \rho_\Lambda - \lambda v^4 \Lambda^4$ (choose $\lambda v^4 \Lambda^4 \ll \rho_\Lambda$)

This contributes at most 0.1% of the dark energy density. The mass and frequency are unaffected — they depend only on lambda Lambda². The fine-tuning of rho_0 is the standard cosmological constant problem, now separated from the torsional sector. This is not a solution to the cosmological constant problem; it is a consistent parameterization.

6. Thermal Coherence and Early Structure

The effective mass is temperature-dependent: $m_\phi(T)$ grows at high T, making the coherence length $\xi(T) = 1/m_\phi(T)$ short. At z ~ 8.2, xi is short enough to favour rapid, compact nucleation — the first 'ice crystals' in the supercooled vacuum. These are the Little Red Dots: regions of incipient torsion alignment that create local mass excess, seeding anomalously early galaxy formation. At $z_t = 1.985$ (thermal transition), the vacuum crystallises globally. At z = 1.087, the gravitational slip reaches its minimum (V-curve, Series A). The DESI 2024 measurement of $f\sigma_8(z)$ should show a feature near z ~ 1.09 that is absent from pure ΛCDM predictions.

7. Observational Predictions and Kill Conditions — 'How to Fire the Theory'

Observable	Prediction	Kill condition	Source
PTA narrow bump	18.52 nHz; sigma=3.70 nHz; alpha>0.01	No feature at 5 sigma in 15-yr data	NANOGrav doi:10.3847/2041-8213/acdac6
DESI fσ8(z)	Feature near z~1.98	No deviation from ΛCDM	doi:10.48550/arXiv.2404.03002
JWST LRDs	Excess at 1.73 Mpc clustering	No excess at 5 sigma	doi:10.1038/s41586-023-05786-2
LiteBIRD biref.	β > 0.05°	β < 0.01°	LiteBIRD 2030
Nanodiamond	85 ppm at 23.93-hr period	No modulation at 5 sigma	Series I (in prep)

Table 1: Kill conditions. If the narrow PTA bump at 18.52 nHz is absent at 5 sigma in 15-year PTA data, the MSDS vacuum lattice is falsified. The frequency cannot be adjusted — it is the only value permitted by $v^2 = 0.00017$.

SERIES D — THE SMOKING GUN

f_c = 18.52 nHz — DERIVED, not chosen

The only frequency allowed by $v^2 = 0.00017$ [Series A doi:10.5281/zenodo.19443940]

sigma = 3.70 nHz (Q=5, 8.6-yr integration window)

T_c ~ mu_CW = 4.187 meV (Coleman-Weinberg scale)

Q_eff ~ 6x10^11 from parametric resonance (decaying inflaton)

lambda ~ 2.9e-98 (fine-tuning stated explicitly)

Kill condition: no narrow feature at 18.52 nHz

If it is there: the vacuum is a crystal

If it is not: MSDS is falsified

The data will decide

Appendix A: Link to the Nanodiamond Sidereal Modulation (Series I)

The vacuum lattice has an anisotropic shear modulus with a preferred axis set by the torsion offset. As the Earth rotates, laboratory experiments coupling to the lattice experience a sidereal modulation. For the forbidden (222) reflection in nanodiamonds (Series I):

Delta-I/I = $v^2/2$ = 0.00017/2 = 8.5x10^-5 = 85 ppm at T = 23.93 hr

Direct laboratory test of the same torsion field producing the 18.52 nHz phonon

Appendix B: Sažetak na bosanskom (Za Dervisovu porodicu)

Naslov: Lokalizovani torsioni mod na 18.52 nHz u gravitaciono-talasnoj pozadini Sažetak: Ovaj rad predvida uski spektralni vrh na frekvenciji 18.52 nHz u gravitacionim talasima koje detektuju pulsar satovi. Ta frekvencija je 'otkucaj srca' vakuumske resetke — ista ona koja oblikuje galaksije i stvara 1.73 Mpc uzorak u rasporedu galaksija. Frekvencija nije izabrana; ona je jedina koja je moguca uz $v^2 = 0.00017$. Laboratorijski test: dijamantska nano-cestica ce pokazati 85 ppm modulaciju sa periodom od 23.93 sata (zvjezdani dan) ako se vakuumska resetka rotira zajedno sa Zemljom. Ako se ovaj signal otkrije, to ce biti prvi direktni dokaz da vakuum nije praznina, vec kristalna struktura. Papiri A, B i C su izgradili osnovu; Papir D povlaci obarac.

Papers A, B, and C built the gun. They established the vacuum lattice, the phase transition, the baryogenesis mechanism. Paper D pulls the trigger. The number is 18.52 nHz. It is not a guess. It is not a fit. It is the Compton frequency of a dilaton condensate with mass 7.66×10^{23} eV, derived from the two anchors of the dual-scale hierarchy. The 8.6-year NANOGrav dataset either contains this feature or it does not. There is no parameter to adjust. The vacuum lattice is either real or it is not. The data will decide.

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